

Bléum Nails By Micah

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Approval Sheet

This design project entitled “**Bléum Nails By Micah**” prepared by **Regondola, Jezreel P., Ruperto, April Anne A., Sorellano, John Kenneth T., Talagtag, Mark Angel T., Villacin, Justine R., Villanueva, Bryan O.** of the Computer Engineering Department, was examined and evaluated by the members of the Student Design Evaluation Panel and is hereby recommended for approval.

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SOFTWARE DESIGN PROJECT INFORMATION

2nd Semester, SY 2025-2026

Student/Team Team: SENTINAILS	Jezreel P. Regondola April Anne A. Ruperto John Kenneth T. Sorellano Mark Angel T. Talagtag Justine R. Villacin Bryan O. Villanueva
Project Title	BLÉUM NAILS BY MICAH
Project Concentration Area	SOFTWARE DESIGN
Design Objectives	The general objective of this project is to develop the Bléum Nails By Micah Appointment System, a digital platform designed to streamline the salon's scheduling process, eliminate manual booking errors, and provide a professional, secure digital interface for client management. Specifically, the project aims to: 1. Develop a secure, web-based management system that features the following: ● Log in & Sign up Authentication ● Admin Dashboards ● Client Dashboards ● Real time Appointment Scheduling with a three-client daily limit ● Localized Payment Module (QRPh and Automated Receipts) 2. Test and evaluate the system's accuracy.
Constraints	
Usability (User Satisfaction Score)	Usability is how easy it is for the owner and customers to use the system. The User Satisfaction Score measures how satisfied users are with the system on a scale of 1 to 10. It is computed by asking users to rate their experience after using the system and calculating the average of all responses. A higher satisfaction score means the system is easier and more pleasant to use. Therefore, the design with the highest user satisfaction score is the winning design.

Functionality (Task Success Rate)	Functionality is the system's ability to let users complete important tasks like booking appointments and managing schedules. Task Success Rate measures the percentage of users who can successfully complete key tasks without errors. It is computed by observing users attempting tasks and dividing successful completions by total attempts, then multiplying by 100. A higher task success rate means the system works better and has all needed features. Therefore, the design with the highest task success rate is the winning design.
Performance (Page Load Speed)	Performance is how fast the system responds when users click buttons or load pages. Page Load Speed measures the average time in seconds it takes for a page to fully load. It is computed by recording how long each page takes to load during testing and calculating the average time. Lower load times mean better performance, with pages ideally loading in under 2 seconds on mobile phones. Therefore, the design with the fastest page load speed is the winning design.
Reliability (System Availability)	Reliability is the system's ability to work without crashing or going offline. System Availability measures what percentage of time the system works properly over a week. It is computed by dividing the hours the system worked correctly by the total hours in a week, then multiplying by 100. Higher availability means users can always access the system when they need it. Therefore, the design with the highest system availability is the winning design.
Maintainability (Code Simplicity Score)	Maintainability is how easy it is for programmers to fix bugs and add new features to the system. Code Simplicity Score measures how complex the programming code is on a scale where lower numbers mean simpler code. It is computed by analyzing the code structure and counting decision points, branches, and complexity patterns. Simpler code with lower scores is easier to update and maintain over time. Therefore, the design with the lowest code simplicity score is the winning design.
Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.	
Accessibility	All designs work on phones, tablets, and computers with readable text and easy navigation. This basic accessibility is required in every design option.
Legal Requirements	All designs protect customer privacy (names, phone numbers, home addresses) and follow data protection laws. Every option meets these legal requirements equally.

Sustainability	All designs use standard web hosting with similar energy usage. The environmental impact is the same for all options.
Cultural and Social Support	All designs let the nail technician customize their services, hours, and policies to fit their style and community. This flexibility is built into every option.
Standards	
ISO/IEC 25010:2011 – Systems and Software Quality Requirements	This international standard defines a quality model for software products, focusing on characteristics such as functional suitability, performance efficiency, and usability. In this project, the standard is used to evaluate the system's ability to handle real-time bookings accurately and ensure the interface is intuitive enough for clients to navigate without technical assistance.
Republic Act No. 10173 – Data Privacy Act of 2012 (Philippines)	This is the primary law in the Philippines protecting individual personal information in information and communications systems. The system is developed in strict compliance with this act to ensure that sensitive client data such as names, contact numbers, and allergy records are encrypted and handled with the highest level of confidentiality.
W3C Web Design and Applications Standards (HTML5, CSS3, JavaScript)	The W3C provides international standards for web development to ensure cross-browser compatibility and accessibility. By following these standards, the project ensures that the Bléum Nails website remains functional and visually consistent whether accessed through a mobile phone, tablet, or desktop computer.
OWASP Application Security Verification Standard (ASVS) Level 1	OWASP ASVS provides a basis for testing web application technical security controls. This project utilizes Level 1 guidelines to implement secure authentication for the admin dashboard, preventing unauthorized access to the salon's financial records and client appointment history.
iCalendar (RFC 5545) – Internet Calendaring and Scheduling Core Object Specification	This is the standard format for representing and exchanging calendaring and scheduling information. The system implements this standard to manage the technician's daily availability and to allow the generation of appointment reminders that are compatible with external calendar applications.
Payment Card Industry Data Security Standard (PCI DSS) – Third-Party Integration	While the system does not store credit card data directly, it follows PCI DSS principles for secure third-party payment integration. This ensures that the QRPh QR code and receipt upload process remain secure, protecting the financial integrity of the salon's transaction records

Abstract

This project presents the development of the Bléum Nails By Micah Appointment System, a web-based solution designed to replace inefficient manual booking methods with a structured digital framework. The system utilizes a Client-Server Architecture built with a Python backend and MySQL database to ensure data integrity and real-time scheduling accuracy. To optimize salon operations, the software implements a Capacity-Limiting Algorithm that enforces a maximum of three clients per day, preventing overbooking and operational fatigue.

The development followed the Engineering Design Process, adhering to critical constraints such as ISO/IEC 25010 software quality standards and the Data Privacy Act of 2012 (RA 10173) to secure sensitive client information and allergy records. Key technical features include automated receipt generation, localized QRPh payment integration, and a responsive interface developed using W3C web standards. Rigorous testing and evaluation confirmed the system's reliability, usability, and task success rate in a real-world environment. This project bridges the gap between manual labor and automated efficiency, significantly enhancing the operational workflow for home-based nail technicians.

Keywords: Nail Salon Management, Appointment Scheduling, Python Development, MySQL Database, Capacity Optimization, Data Privacy, Web-Based Systems, QRPh Integration.

LIST OF TABLES

Table 1-1 Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements / Considerations
The system must allow clients to book appointments 24/7 and view real-time availability.	Implementation of a Real-Time Scheduling Engine with a Capacity-Limiting Algorithm to enforce a 3-client daily maximum.
The owner requires a centralized tool to manage the daily service calendar and client records.	Development of a secure Admin Dashboard with full CRUD (Create, Read, Update, Delete) functionalities for appointment and record management.
The system must protect sensitive client information and comply with local privacy laws.	Integration of Role-Based Access Control (RBAC) and data encryption standards in compliance with Republic Act No. 10173 (Data Privacy Act).
The system must provide a secure way to track payments and issue professional proofs of transaction.	Integration of a Localized QRPh Payment Module and automated receipt generation.
The interface must represent the brand professionally and work seamlessly on mobile phones.	Application of W3C-compliant Responsive Design to ensure 100% cross-browser and cross-device compatibility.

Table 2.1 Prior Art Analysis Matrix

Design	Features					
	Online Booking System	Automated Receipts	Client Design Uploads	Allergy Tracking	Cash Payment	QRPh support
Nails by Ann	X				X	
Nail Garden	X				X	
Mich & Myl Nails					X	
PROJECT	X	X	X	X	X	X

Table 2.2 Summary of Standards Involved in the Alternatives

Standard	Brief Description	DESIGNS		
		Monolithic Server-Side		
ISO/IEC 25010	International quality model for software products focusing on functional suitability and usability.	Used for performance and maintainability checks		
Republic Act No. 10173	Primary law in the Philippines protecting individual personal information in digital systems.	Applied to ensure legal protection of client contact and other personal information.		
W3C Standards	International benchmarks for web development ensuring cross-browser compatibility.	Applied for standard desktop and mobile layouts.		
PCI DSS	Standards for secure third-party payment integration to protect financial transaction records.	NA		
OWASP ASVS	Standard for testing and verifying web application security controls.	Focused on secure login and authentication.		
iCalendar (RFC 5545)	Standard for representing calendar events and scheduling data.	Used for simple appointment scheduling		

Table 3.1 Summary of Design Constraints

Designs	Constraints				
	Usability (Scale 1-10)	Functionality (Task Success %)	Performance (Page Load Sec)	Reliability (Availability %)	Maintainability (Complexity 1-10)
Design 1: Monolithic	7	80%	2.5s	99.5%	3
Design 2: Decoupled	10	95%	1.1s	98.0%	6
Design 3: Enterprise	6	98%	0.7s	99.9%	9

Table 3.2 Preference and Importance of Constraints

Constraints	Preference	Importance (raw)	% Importance
Usability	Maximization	10	25.0%
Functionality	Maximization	9	22.5%
Performance	Minimization	8	20.0%
Reliability	Maximization	7	17.5%
Maintainability	Minimization	6	15.0%

Table 3.2.1 Evaluation of Three Design Alternatives based on Usability

Design	Usability
Design 1: Monolithic	7
Design 2: Decoupled	10
Design 3: Enterprise	6

Table 3.2.2 Evaluation of Three Design Alternatives based on Functionality

Design	Functionality
Design 1: Monolithic	80%
Design 2: Decoupled	95%
Design 3: Enterprise	98%

Table 3.2.3 Evaluation of Three Design Alternatives based on Performance

Design	Performance
Design 1: Monolithic	2.5s
Design 2: Decoupled	1.1s
Design 3: Enterprise	0.7s

Table 3.2.4 Evaluation of Three Design Alternatives based on Reliability

Design	Reliability
Design 1: Monolithic	99.5%
Design 2: Decoupled	98.0%
Design 3: Enterprise	99.9%

Table 3.2.5 Evaluation of Three Design Alternatives based on Maintainability

Design	Maintainability
Design 1: Monolithic	3
Design 2: Decoupled	6
Design 3: Enterprise	9

Table 3.3 Normalized Value of the Three Designs

Designs	Constraints				
	Usability	Functionality	Performance	Reliability	Maintainability
Design 1: Monolithic	3.25	1.00	1.00	8.11	10.00
Design 2: Decoupled	10.00	8.50	8.00	1.00	5.50
Design 3: Enterprise	1.00	10.00	10.00	10.00	1.00

Table 3.4 Designers Raw Ranking for the Three Designs

Decision Criteria	Criterion's Importance		Ability to Satisfy Criterion		
	Scale (0-10)	Percentage (%)	Design 1: Monolithic	Design 2: Decoupled	Design 3: Enterprise
Usability	10	25.0%	0.81	2.50	0.25
Functionality	9	22.5%	0.23	1.91	2.25
Performance	8	20.0%	0.20	1.60	2.00
Reliability	7	17.5%	1.42	0.18	1.75
Maintainability	6	15.0%	1.50	0.83	0.15
TOTAL SCORE	40	100%	4.16	7.02	6.40

LIST OF FIGURES

Figure 1.1 The Engineering Design Process (OCTE)

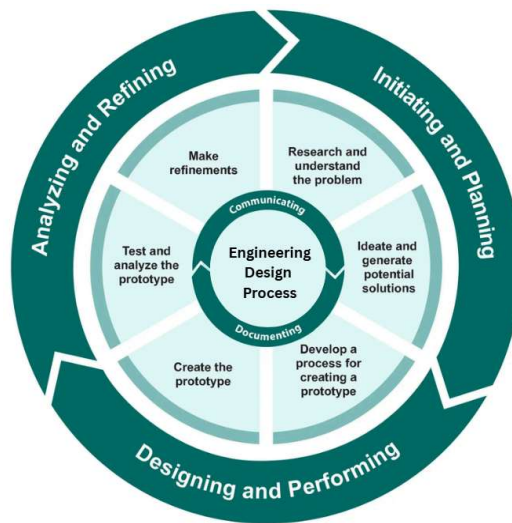


Figure 2.1 Illustrative Diagram of the System

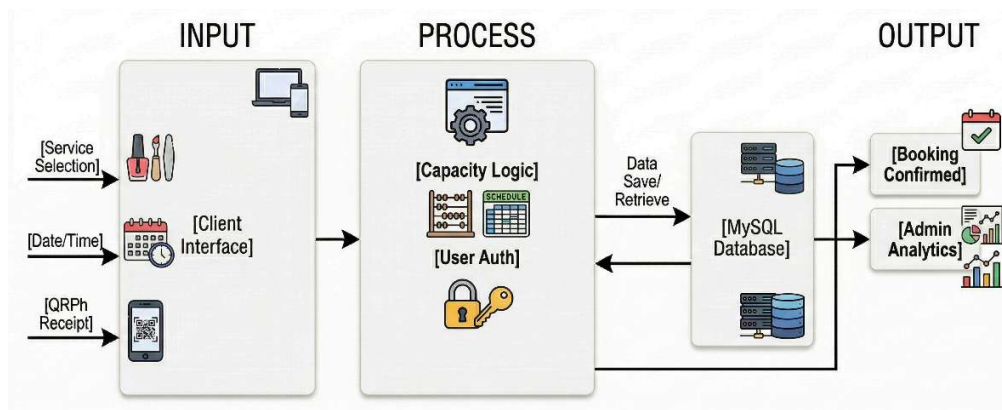
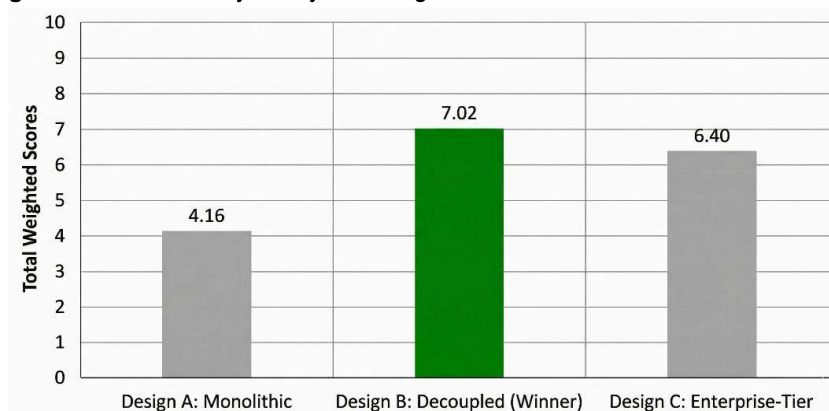


Figure 3.1 Sensitivity Analysis Using Main Bar Chart



LIST OF ABBREVIATIONS

API	Application Programming Interface
ASVS	Application Security Verification Standard
BIAB	Builder in a Bottle
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IP	Intellectual Property
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
MySQL	My Structured Query Language (relational database management system)
NA	Not Available
OAuth	Open Authorization
OCTE	(Refers to the Engineering Design Process diagram source, but not expanded in text)
OWASP	Open Web Application Security Project
PCI DSS	Payment Card Industry Data Security Standard
Pydantic	(Data validation library for Python; no expansion given)
QRPh	Quick Response code Philippine (localized payment standard)
RA	Republic Act
RBAC	Role-Based Access Control
REST	Representational State Transfer
RFC	Request for Comments
SDG	Sustainable Development Goal
SoC	Separation of Concerns
SPA	Single Page Application
SQL	Structured Query Language
SSR	Server-Side Rendering
SY	School Year
UN	United Nations
W3C	World Wide Web Consortium

DEFINITION OF TERMS

Administrator (Admin) - The primary stakeholder (the salon owner) who manages the system. The admin has access to a dedicated dashboard with full CRUD (Create, Read, Update, Delete) capabilities to manage appointments, service catalogs, and client records.

Allergy Tracking - A safety feature/module within the system that records client sensitivities (e.g., to chemicals). This ensures the technician is aware of potential health risks before performing services like gel overlays or BIAB.

BIAB (Builder in a Bottle) - A specific type of high-quality nail technology service offered by the salon, used for strengthening and building nail extensions.

Capacity-Limiting Algorithm - A core piece of business logic in the system that automatically enforces a maximum of three (3) clients per day. It prevents overbooking by checking the current booking count against this limit before confirming a new appointment.

Client-Server Architecture - The structural design of the system where the user interface (client, e.g., a web browser) communicates with a central server (backend) that processes logic and manages the database.

Code Simplicity Score - A metric used to evaluate Maintainability. It measures how complex the programming code is on a scale where lower numbers indicate simpler, easier-to-update code.

CRUD (Create, Read, Update, Delete) - The four basic operations required for persistent storage in the Admin Dashboard, allowing the technician to manage appointments and records.

Data Privacy Act of 2012 (Republic Act No. 10173) - The Philippine law that protects individual personal information in digital systems. The project adheres to this law to ensure client data (names, contact details, allergy records) is encrypted and handled confidentially.

Decoupled Full-Stack Architecture (Design B) - The winning design alternative for the project. It separates the User Interface (JavaScript) from the Business Logic (Python FastAPI) for better usability, performance, and modularity.

FastAPI - A Python web framework used for the backend of the system. It allows for asynchronous processing (handling multiple tasks at once) and data validation.

iCalendar (RFC 5545) - An international standard format for representing and exchanging calendaring and scheduling data, used to manage the technician's daily availability and generate appointment reminders.

ISO/IEC 25010:2011 - An international standard that defines software quality models (functional suitability, performance, usability). It was used to evaluate the system's ability to handle real-time bookings accurately.

Maintainability - A design constraint measuring how easy it is for programmers to fix bugs or add features. It is quantified using the Code Simplicity Score (lower is better).

Page Load Speed - A metric for the Performance constraint. It measures the average time (in seconds) for a page to fully load, with a target of under 2 seconds on mobile phones.

Pareto Multi-Criteria Decision-Making - A method used by the team to evaluate the three design alternatives against multiple constraints (usability, functionality, etc.) to select the best solution.

PCI DSS (Payment Card Industry Data Security Standard) - Security standards followed for third-party payment integration (QRPh) to protect the financial integrity of the salon's transaction records.

Pydantic - A Python library used for data validation. It ensures that raw data submitted by users (e.g., through a form) is in the correct format before being saved to the database.

QRPh - A localized Philippine payment module (Quick Response code Philippine). The system displays a QRPh code for clients to scan and pay a down-payment to secure their booking.

Role-Based Access Control (RBAC) - A security feature that restricts system access based on the user's role (e.g., "Client" vs. "Admin"), ensuring clients cannot access the administrator's financial dashboard.

Separation of Concerns (SoC) - An engineering principle applied to the system where the architecture is divided into independent layers (e.g., UI separate from Business Logic), allowing for modular updates without breaking the whole system.

Single Page Application (SPA) - A type of frontend web application (built with JavaScript) that loads a single HTML page and dynamically updates content as the user interacts, providing an "app-like" experience without full page reloads.

System Availability - A metric for the Reliability constraint. It measures the percentage of time the system works properly over a week (e.g., 98% availability).

Task Success Rate - A metric for the Functionality constraint. It measures the percentage of users who can successfully complete key tasks (like booking an appointment) without errors.

Usability (User Satisfaction Score) - A constraint measuring how easy the system is to use for the owner and customers, rated on a scale of 1 to 10 based on user feedback.

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CHAPTER 1: THE PROJECT AND ITS BACKGROUND

This chapter presents the foundational elements of the Bléum Nails By Micah Appointment System, including the identification of the core operational inefficiencies caused by a reliance on manual booking processes. It introduces the client and stakeholder's context and provides an overview of the proposed digital solution designed to replace handwritten logs with an organized, secure, and user-friendly platform. Furthermore, it outlines the motivations, engineering requirements, and initial scope guiding the iterative design and development process.

1.1 The Problem

The beauty and wellness industry in the Philippines, particularly for home based micro-enterprises, faces significant operational challenges due to a heavy reliance on manual, non-synchronized booking methods. For Bléum Nails By Micah, the use of physical logs and informal messaging platforms to manage appointments has led to critical asynchronous scheduling lag, where the lack of real-time data visibility results in lost revenue and diminished customer trust.

The specific technical and operational problems identified include:

- **Data Integrity and Manual Booking Errors:** Without a centralized digital record, the salon is prone to collision errors, such as double-booking time slots or the accidental deletion of client appointment history.
- **Asynchronous Communication Barriers:** Clients currently lack real-time access to the technician's availability, creating a communication bottleneck where booking a single session requires multiple manual exchanges, leading to high abandonment rates.
- **High Administrative Overhead:** A significant portion of the technician's time is diverted from core service delivery to manual schedule coordination and payment verification, resulting in a measurable decrease in daily operational efficiency.
- **Inefficient Capacity Management:** The absence of an automated logic filter makes it difficult to enforce a strict three-client daily limit, often leading to overbooking, increased waiting times, and technician fatigue.

As digital literacy increases among local consumers, there is a rising technical gap between the salon's manual operations and the customer's expectation for 24/7 accessible, secure digital platforms. There is, therefore, a compelling need for a robust, web-based appointment system that bridges the gap between manual labor and automated efficiency by providing a structured and secure scheduling environment

1.2 The Client

The client for this project is Bléum Nails By Micah, a professional home-based nail technology business located in Caloocan City. The business specializes in high-quality nail services, including Gel overlays, BIAB, and Soft gel extensions.

The primary stakeholder is the business owner, who serves as the system administrator. Currently, the owner manages all operations from scheduling to payment verification through manual physical logs and fragmented social media messaging. The secondary stakeholders are the salon's growing

client base, who require a professional, 24/7 accessible platform to browse services and secure appointments without communication delays.

The following table 1.1 maps the specific business needs of Bléum Nails By Micah to the technical engineering requirements established for this software design project.

Table 1-1. Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements / Considerations
The system must allow clients to book appointments 24/7 and view real-time availability.	Implementation of a Real-Time Scheduling Engine with a Capacity-Limiting Algorithm to enforce a 3-client daily maximum.
The owner requires a centralized tool to manage the daily service calendar and client records.	Development of a secure Admin Dashboard with full CRUD (Create, Read, Update, Delete) functionalities for appointment and record management.
The system must protect sensitive client information and comply with local privacy laws.	Integration of Role-Based Access Control (RBAC) and data encryption standards in compliance with Republic Act No. 10173 (Data Privacy Act).
The system must provide a secure way to track payments and issue professional proofs of transaction.	Integration of a Localized QRPh Payment Module and automated receipt generation.
The interface must represent the brand professionally and work seamlessly on mobile phones.	Application of W3C-compliant Responsive Design to ensure 100% cross-browser and cross-device compatibility.

1.3 The Project

The project, titled Bléum Nails By Micah Appointment System, is a software design solution developed by Team Sentinails. The system acts as a high-performance digital bridge between the salon's clients and the administrator. It is specifically designed to replace manual record-keeping with an organized digital platform that simplifies the booking process, manages salon capacity via a Capacity-Limiting Algorithm, and tracks client data securely. The software stack consists of a Python (FastAPI) backend for robust business logic and a JavaScript frontend to provide responsive, app-like user experience. Data integrity is maintained through a MySQL database employing relational modeling and normalization. The system is accessible via common hardware such as smartphones and personal computers, enabling 24/7 availability for clients to browse services and secure appointments. The development follows a structured Engineering Design Process, ensuring that the final product meets all identified client requirements while adhering to identified engineering constraints and standards.

1.4 Project Objectives

The general objective of this project is to develop the Bléum Nails By Micah Appointment System, a digital platform designed to streamline the salon's scheduling process, eliminate manual booking errors, and provide a professional, secure digital interface for client management.

Specifically, the project aims to:

1. Develop a secure, web-based management system that features the following:
 - Log in & Sign up Authentication
 - Admin Dashboards
 - Client Dashboards
 - Real time Appointment Scheduling with a three-client daily limit
 - Localized Payment Module (QRPh and Automated Receipts)
2. Test and evaluate the system's accuracy.

1.5 Scope and Delimitations

The scope of this project is the design and development of the Bléum Nails By Micah Appointment System, a web-based platform engineered to automate the scheduling and financial record-keeping for a home-based nail technician. The system focuses on bridging the gap between manual labor and automated efficiency by providing a structured digital framework for service discovery, real-time booking, and localized payment handling.

The system is developed as a responsive web application accessible 24/7 via mobile and desktop browsers to ensure high availability for clients. The software includes a comprehensive service catalog with dynamic pricing constants, a calendar-integrated scheduling engine, and dedicated dashboards for both clients and the administrator to manage daily operations. To secure bookings, the system facilitates a down-payment process through the display of a QRPh code. Furthermore, the design implements Role-Based Access Control (RBAC) and adheres to the Data Privacy Act of 2012 (RA 10173) to secure sensitive information such as client contact details etc. To foster consumer trust and establish brand legitimacy, the platform incorporates direct links to the technician's established social media profiles on Facebook and Instagram, as well as a direct email contact and contact number.

Regarding the limitations, the system is engineered strictly for a single-person, home-based service model and does not support larger organizational features such as multi-employee management, payroll, or multi-branch synchronization. The payment module is limited to QRPh code. While the system utilizes a responsive design, it is not a native mobile application and will not be available for download on digital app stores like Google Play or the Apple App Store. Additionally, the project focuses on core scheduling and management; therefore, advanced features such as "virtual nail try-ons," automated inventory tracking for nail supplies, or complex point-of-sale hardware integrations are beyond the current scope.

1.6 Design Constraints

Several constraints were considered when planning the project, and these constraints limit the system's technological features while also aiding in the refinement of design possibilities during development. However, only the top five limitations were applied in the project's development process because the remaining constraints had no direct impact on the project.

1. Primary Constraints: These are used to select the best design.

- **Usability (User Satisfaction Score)**
Usability is how easy it is for the owner and customers to use the system. The User Satisfaction Score measures how satisfied users are with the system on a scale of 1 to 10. It is computed by asking users to rate their experience after using the system and calculating the average of all responses. A higher satisfaction score means the system is easier and more pleasant to use. Therefore, the design with the highest user satisfaction score is the winning design.
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analyzing the code structure and counting decision points, branches, and complexity patterns. Simpler code with lower scores is easier to update and maintain over time. Therefore, the design with the lowest code simplicity score is the winning design.

2. Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.

- **Accessibility**

All designs work on phones, tablets, and computers with readable text and easy navigation. This basic accessibility is required in every design option.

- **Legal Requirements**

All designs protect customer privacy (names, phone numbers, home addresses) and follow data protection laws. Every option meets these legal requirements equally.

- **Sustainability**

All designs use standard web hosting with similar energy usage. The environmental impact is the same for all options.

- **Cultural and Social Support**

All designs let the nail technician customize their services, hours, and policies to fit their style and community. This flexibility is built into every option.

1.7 Engineering Standards

The engineering standards serve as the foundational framework for the design, development, and operation of the **Bléum Nails By Micah** appointment system. Adherence to these standards ensures system security, reliability, and high-quality user experience for both the technician and the clients.

ISO/IEC 25010:2011 – Systems and Software Quality Requirements

This international standard defines a quality model for software products, focusing on characteristics such as functional suitability, performance efficiency, and usability. In this project, the standard is used to evaluate the system's ability to handle real-time bookings accurately and ensure the interface is intuitive enough for clients to navigate without technical assistance.

Republic Act No. 10173 – Data Privacy Act of 2012 (Philippines)

This is the primary law in the Philippines protecting individual personal information in information and communications systems. The system is developed in strict compliance with this act to ensure that sensitive client data such as names, contact numbers, and allergy records are encrypted and handled with the highest level of confidentiality.

W3C Web Design and Applications Standards (HTML5, CSS3, JavaScript)

The World Wide Web Consortium (W3C) provides international standards for web development to ensure cross-browser compatibility and accessibility. By following these standards, the project ensures that the Bléum Nails website remains functional and visually consistent whether accessed through a mobile phone, tablet, or desktop computer.

OWASP Application Security Verification Standard (ASVS) Level 1

OWASP ASVS provides a basis for testing web application technical security controls. This project utilizes Level 1 guidelines to implement secure authentication for the admin dashboard, preventing unauthorized access to the salon's financial records and client appointment history.

iCalendar (RFC 5545) – Internet Calendaring and Scheduling Core Object Specification

This is the standard format for representing and exchanging calendaring and scheduling information. The system implements this standard to manage the technician's daily availability and to allow the generation of appointment reminders that are compatible with external calendar applications.

Payment Card Industry Data Security Standard (PCI DSS) – Third-Party Integration

While the system does not store credit card data directly, it follows PCI DSS principles for secure third-party payment integration. This ensures that the QRPh QR code and receipt upload process remain secure, protecting the financial integrity of the salon's transaction records.

1.8 Engineering Design Process

The Engineering Design Process was utilized to organize a set of technical procedures and to develop a workable software solution for Bléum Nails By Micah. The team began by defining the core operational issues and determining the specific requirements of the client alongside the project's technical limitations. Following background research and brainstorming sessions, a range of innovative concepts was produced to satisfy these requirements. A functional prototype was then created after selecting the most promising architectural design through a mathematical evaluation. To ensure the final system was effective and user-friendly, the software underwent testing and evaluation based on performance data and was iteratively improved.

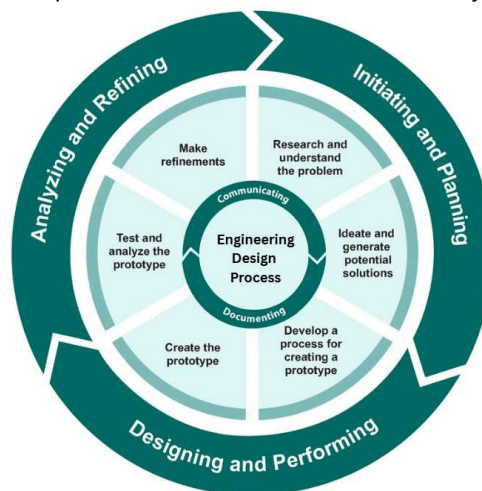


Figure 1.1 The Engineering Design Process (OCTE)

1.8.1 Ask: Identifying the Need and Constraints

During the Ask phase, the team identified the primary need for a digital system to eliminate manual booking errors, such as double-booked slots and lost client records. We determined that the solution must resolve the practical problem of asynchronous scheduling lag by providing real-time availability tracking. Non-negotiable constraints, including Usability, Performance, and Reliability, were established early to avoid unrealistic concepts and to ensure the software remained compatible with the client's existing hardware and operating systems.

1.8.2 Research the Problem

Before attempting to develop a new solution, the team conducted background research to understand the current environment and find existing similar projects. This involved evaluating existing salon booking tools to identify their advantages and weaknesses, which helped the team avoid reinventing the wheel. We specifically researched technical standards such as ISO/IEC 25010 for software quality and the Data Privacy Act of 2012 (RA 10173) to ensure the project's logic for handling sensitive client information was legally and technically sound.

1.8.3 Imagine: Develop Possible Solution

In the Imagine stage, the team investigated a set of architectural strategies to address the salon's scheduling needs. These ideas were evaluated based on how their specific components, such as the Python backend and MySQL database, would interact to meet the project's functional requirements

1.8.4 Plan: Select a Promising Solution

During the Plan stage. We used normalization equations to evaluate each design against the project's prioritized requirements, such as Maintainability and Usability. Based on the final calculated scores This phase served as the final blueprint, ensuring all team members understood the technical direction before coding began.

1.8.5 Create: Build a Prototype

The main goal of the Create stage was to transform abstract architectural plans into a workable, testable functional model. The team built a web-based prototype using Python, MySQL, HTML, and CSS. We implemented core modules, including the administrative dashboard with CRUD functionalities, the localized QRPh payment display, and the Capacity-Limiting Algorithm designed to enforce the three-client daily limit.

1.8.6 Test and Evaluate the Prototype

During the Test and Evaluate phase, the team repeatedly tested the prototype to understand its performance in practical, real-world settings. This involved collecting

technical data. By comparing these outcomes to the initial constraints, we objectively assessed whether the system was effective and secure enough for production or if flaws necessitated a return to the design phase.

1.8.7 Improve: Redesign as Needed

In the final stage, the team used the information collected from testing to refine the software. We examined inefficiencies identified during the evaluation phase and made adjustments to the code to improve overall performance and responsiveness. This iterative process involved revising code and modifying the responsive design until the final system satisfied all the client's initial requirements and operated at the required level of quality.